CHAPTER WISE PUBLIC 2 MARK QUESTIONS – 2020 TO 2025

1. ELECTROSTATICS

1. What is corona discharge? **(MARCH 2020, MAY 2022)**
2. Potential in a given region is given as a function of distance x, $V=5(x^2+x)$ Volt. Find the electric field when x=1 cm. **(SEP 2020)**
3. Define "Electrostatic Potential". **(SEP 2021, MAR 2024)**
4. Dielectric strength of air is 3×10^6 V m^{-1}. Suppose the radius of a hollow sphere in the Van de Graff generator is R = 0.5 m, calculate the maximum potential difference created by this Van de Graaff generator. **(SEP 2021)**
5. State Gauss law. **(SEP 2022)**
6. Define electric field' (**MAR 2023)**
7. Define Capacitance (**JUNE 2023)**
8. A sample of HCL gas is placed in a uniform electric field of magnitude 3×10^4 N/C. The dipole moment of each HCL molecule is 3.4×10^{-30} Cm . Calculate the maximum torque experienced by each HCL molecule. (**JUNE 2024)**
9. A parallel plate capacitor has two square plates of side 5 cm and separated by a distance of 1 mm. Calculate the capacitance of this capacitor. (**MARCH 2025)**
10. Calculate the ratio of electric potential due to a point charge q at a point which is at a distance r to the electric potential at a point when the distance is doubled. (**JULY 2025)**

ABS PHYSICS

2. CURRENT ELECTRICITY

1. If an electric field of magnitude 570 NC^{-1}, is applied in the copper wire, find the acceleration experienced by the electron. **(SEP 2020)**
2. What is Peltier effect ? **(SEP 2021)**
3. Define electrical resistivity. **(MAY 2022)**
4. Calculate the equivalent resistance for the circuit where 4 Ω and 2 Ω resistors are connected in series to 12 V battery And also find the potential difference across 2 Ω and 4 Ω resistors in the circuit. **(SEP 2022)**
5. If the resistance of coil is 3.2 **Ω** at 20°C and a= 0.004/°C then, determine its resistance at 100°C. (**MAR 2023)**
6. Determine the number of electrons flowing per second through a conductor when a current of 32 A flows through it. (**JUNE 2023)**

7. State the applications of Seebeck Effect. **(MARCH 2024)**

8. A copper wire of cross-sectional area 10^{-6} mm² carries a current of 2 A. Calculate the current density. **(JUNE 2024)**

9. A cell supplies a current of 0.9 A through a 20 Ω resistor and a current of 0.3 A through a 72 Ω resistor. Calculate the internal resistance of the cell **(MARCH 2025)**

10. Distinguish between drift velocity and mobility. (**JULY 2025)**

3. MAGNETISM & MAGNETIC EFFECT OF ELECTRIC CURRENT

1. Two materials X and Y are magnetised whose intensity of magnetisation are 500 Am^{-1} and 2000 Am^{-1} respectively. The magnetising field is 1000 Am^{-1}. What is the ratio between the susceptibilities of the two material? **(MARCH 2020)**

2. State Ampere's Circuital Law. **(SEP 2021, MAR 2023 , JUNE 2024)**
3. How will you increase the current sensitivity of a galvanometer ? **(MAY 2022)**
4. Define ampere in terms of force. **(SEP 2022)**
5. State Flemings left hand rule (**JUNE 2023)**
6. What is meant by hysteresis? **(MARCH 2024)**
7. Write the limitations of Cyclotron. **(MARCH 2025)**

8. Is an ammeter connected in series or parallel in a circuit? Why? **(JULY 2025)**

4. ELECTROMAGNETIC INDUCTION & ALTERNATING CURRENT

1. An ideal transformer has 460 and 40,000 turns in the primary and secondary coils respectively. Find the voltage developed per turn of the secondary coil if the transformer is connected to a 230 V AC main. **(MARCH 2020)**
2. State Lenz's law. (**SEP 2020 , JUNE 2023, MARCH 2025)**
3. Mention the ways of producing induced emf. **(SEP 2021, SEP 2022**, **JUNE 2024)**
4. State Fleming's right hand rule. **(MAY 2022, JULY 2025)**

5. How will you define Q-factor ? (**MAR 2023)**

6. A straight metal wire crosses a magnetic field of flux 4 mWb in a time 0.4 sec. Find the magnitude of the emf induced in the wire. **(MARCH 2024)**

ABS PHYSICS

5. ELECTROMAGNETIC WAVES

1. What are the uses of X-rays ? **(MARCH 2020 , SEP 2020)**
2. Give two uses of IR radiation. **(SEP 2021, JUNE 2024)**
3. What is displacement current ? **(MAY 2022)**
4. Give two uses of UV radiation.. **(SEP 2022)**
5. What are electromagnetic waves ? **(MARCH 2024)**

6. RAY OPTICS

1. What are paraxial rays and marginal rays? **(SEP 2020)**
2. Why does sky appear blue ? **(SEP 2021 JUNE 2024)**
3. The angle of minimum deviation for the equilateral prism is 40°. Find the refractive index of the material of the prism. **(MAY 2022)**
4. Pure water has refractive index 1.33. What is the speed of light through it ? **(SEP 2022)**
5. Explain the reason for the glittering of diamond. (**MAR 2023)**
6. What is the reason for reddish appearance of sky during sunrise and sunset? (**JUNE 2023)**
7. If the focal length is 150 cm for a lens, what is the power of the lens ? **(MARCH 2024)**
8. What is optical path ? **(MARCH 2025)**
9. Light travelling through transparent oil enters into glass of refractive index 1.5. If the refractive index of glass with respect to the oil is 1.25, what is the refractive index of the oil? **(JULY 2025)**

7. WAVE OPTICS

1. Distinguish between Fresnel and Fraunhofer types of diffraction. **(MARCH 2020)**
2. Find the Polarizing angle for glass of refractive index 1.5. **(SEP 2021)**
3. The ratio of intensities of two waves in an interference pattern is 36 : 1. What is the ratio of the amplitudes of the two interfering waves ? (**MAR 2023)**
4. Find the polarizing angle for a glass of refractive index 1.5. (**JUNE 2023)**
5. State Malus Law. **(MARCH 2024)**
6. What is Double Refraction ? **(JUNE 2024)**
7. What are the shapes of wavefront for a: (a) Source at infinite (b) Point source (c) Line source **(MARCH 2025)**
8. How do source and images behave as coherent sources? Explain in brief. **(JULY 2025)**

ABS PHYSICS

8. DUAL NATURE OF RADIATION & MATTER

1 .Why electron is preferred over X-ray in microscope? **(MARCH 2020)**

2. Calculate the cut-off wavelength and cut-off frequency of X rays from an X ray tube of accelerating Potential 20,000 V. **(SEP 2020, JUNE 2024)**

3. What is Photovoltaic cell? **(SEP 2020)**

4. How will you define threshold frequency ? **(SEP 2021)**

5. Define work function of a metal. Mention its unit. **(MAY 2022 , MAR 2023, JULY 2025)**

6. Define stopping potential. **(SEP 2022, MAR 2025)**

7. What is photoelectric effect? (**JUNE 2023)**

9. ATOMIC & NUCLEAR PHYSICS

1. What are the properties of neutrino? **(MARCH 2020)**

2. What are the constituent particles of Neutron and Proton ? **(SEP 2020)**

3. Calculate the radius of $_{79}Au^{197}$ nucleus. **(MAY 2022)**

4. Define atomic mass unit. **(SEP 2022)**

5. What is meant by activity or decay rate ? Give its unit. (**MAR 2023)**

6. The Radius of the 5 th orbit of hydrogen atom is 13.25 **A°** Calculate the Debroglie Wavelength of electron in the 5 th orbit. (**JUNE 2023)**

7. The half-life of radioactive sample is 5.01 days. Calculate the decay constant. **(MARCH 2024)**

8. Define Ionisation Energy. **(JUNE 2024)**

9. What is mass defect? Give its expression. **(MARCH 2025)**

10. What is meant by excitation energy? **(JULY 2025)**

10. ELECTRONICS & COMMUNICATION SYSTEMS

1. What do you mean by doping? **(MARCH 2020 ,MAY 2022)**

2. What is skip area? **(MARCH 2020)**

3. Define skip distance. **(SEP 2020)**

4. What is rectification ? **(SEP 2022)**

5. Draw the circuit diagram of a full wave rectifier. (**MAR 2023)**

6. Distinguish between Intrinsic and Extrinsic Semiconductor. (**JUNE 2023)**

7. What is meant by biasing? Mention its types. **(MARCH 2024)**

ABS PHYSICS

8.Draw the circuit diagram for a forward biased p-n junction diode. **(MARCH 2025)**

9. Give any two applications of RADAR. **(JULY 2025)**

CHAPTER WISE PUBLIC 3 MARK QUESTIONS – 2020 TO 2025

1. ELECTROSTATICS

1. Derive the expression for equivalent capacitance, when capacitors are connected in parallel. **(MARCH 2020)**
2. Obtain Gauss's law of electrostatics from Coulomb's inverse square law. **(MARCH 2024) (SEP 2020)**
3. Obtain the expression for energy stored in the parallel plate capacitor. **(SEP 2021, 2022)**
4. Derive the expression for resultant capacitance, when capacitors are connected in series. **(MAY 2022)**
5. Derive an expression for electrostatic potential due to a point charge. **(MAR 2023)**
6. Calculate the electric flux through the rectangle of side 5 cm and 10 cm kept in a uniform electricfield of 100 N/C given the angle is 60° . if the angle is zero degree calculate the electric flux. **(JUNE 2023)**
7. Differentiate coulomb force and gravitational force. **(JUNE 2024, MAR 2025)**
8. Derive an expression for the torque experienced by a dipole due to a uniform electric field. **(JULY 2025)**

2. CURRENT ELECTRICITY

1. The resistance of a nichrome wire at 0°C is 10 Ω. If its temperature coefficient of resistance is 0.004/°C, find its resistance at boiling point of water. Comment on the result. **(MARCH 2020)**
2. State Kirchoff's Current and Voltage laws. **(SEP 2020 , SEP 2021, MAR 2023 , MAR 2025)**
3. Obtain a relation between current and drift velocity. **(MAY 2022)**
4. What is Seebeck effect? State the applications of Seebeck effect. **(SEP 2022)**
5. State and explain the principle of potentiometer. **(JUNE 2023)**
6. A copper wire of cross-sectional area 0.5 mm² carries a current of 0.2 A. If the free electron density of copper wire is 8.4×10^{28} m^3, then compute the drift velocity of free electron. **(MARCH 2024)**
7. Calculate the effective resistance when resistors are connected in series. **(JUNE 2024)**
8. The resistance of a nichrome wire at 20°C is 10Ω. If its temperature coefficient of resistivity of nichrome is 0.008/°C, find the resistance of the wire at boiling point of water. Comment on the result. **(JULY 2025)**

3. MAGNETISM & MAGNETIC EFFECT OF ELECTRIC CURRENT

1. Explain the conversion of galvanometer into voltmeter. **(MARCH 2020)**
2. Compare the properties of dia, para and ferromagnetic materials. **(SEP 2020, JULY 2025)**

ABS PHYSICS

3. The repulsive force between two magnetic poles in air is 9×10^{-3} N. If the two poles are equal in strength and are separated by a distance of 10 cm, calculate the pole strength of each pole. **(SEP 2021)**

4. List out salient features of magnetic Lorentz force. **(MAY 2022, MAR 2025)**

5. An electron moving perpendicular to a uniform magnetic field 0.500 T undergoes circular motion of radius 2.50 mm. What is the speed of electron ? **(SEP 2022)**

6. Explain the conversion of galvanometer into ammeter. **(MARCH 2023)**

7. **A coil of tangent galvanometer of diameter 0.24 m has 100 turns. If the horizontal component of Earths magnetic field is 25×10^{-6} T then calculate the cu/rrent which gives deflection of 60°. (JUNE 2023)**

8 . Define current sensitivity. State the factors which increase the current sensitivity of a galvanometer. **(MARCH 2024)**

4. ELECTROMAGNETIC INDUCTION & ALTERNATING CURRENT

1. What are the advantages and disadvantages of AC over DC? **(MARCH 2020, MAR 2025)**

2. An 500 μH inductor, $\frac{80}{\pi^2}$ pF capacitor and a 628 Ω resistor are connected to form a series RLC circuit. Calculate the resonant frequency and Q-factor of this circuit at resonance. **(SEP 2020)**

3. Mention the various energy losses in a transformer. **(SEP 2021, JUNE 2023, JULY 2025)**

4. **Find the impedance of a series RLC circuit, if the inductive reactance, capacitive reactance and resistance are 184 Ω, 144 Ω, and 30 Ω respectively. Also calculate the phase angle between voltage and current. (MAY 2022)** **********

5. AC is advantageous than DC. Explain. **(SEP 2022)**

6. How will you induce an emf by changing the area enclosed by the coil ? **(MARCH 2023)**

7. Derive the equation for inductance of a solenoid. Assume that the length of the solenoid is greater than Its diameter. **(MARCH 2024)**

8. A coil of 200 turns carries a current of 4 *A*. If the magnetic flux through the coil is 6×10^{-5} Wb, find the magnetic energy stored in the medium surrounding the coil. **(EXERCISE PROBLEM) (JUNE 2024)**

5. ELECTROMAGNETIC WAVES

1. What are Fraunhofer lines? How are they useful in the identification of elements present in the Sun ? **(MARCH 2023, JULY 2025)**

2. Write down Maxwell equations in Integral form. **(JUNE 2023)**

3. Write down the properties of electromagnetic waves. **(JUNE 2024, MAR 2025)**

ABS PHYSICS

6. RAY OPTICS

1. **What is total internal reflection? Give the condition for the total internal reflection takes place. (SEP 2020)** **********
2. What are critical angle and total internal reflection ? **(SEP 2021)**
3. Derive the relation between f and R for a spherical mirror. **(MAY 2022 , JUNE 2024)**
4. **Light travels from air into a glass slab of thickness 50 cm and refractive index 1.5. What is the speed of light in the glass slab and what is the time taken by the light to travel through the glass slab ? (SEP 2022)** **********
5. What is optical path ? Write down the equation for optical path and mention what each term represents. **(MAR 2023)**
6. Find the ratio of the intensities of light with wavelength 500 nm and 300 nm which undergo Rayleigh scattering. **(JUNE 2023)**
7. Derive the equation for effective focal length for lenses in contact. **(MARCH 2024)**

7. WAVE OPTICS

1. Two light sources of equal amplitudes interfere with each other. Calculate the ratio of maximum and minimum intensities. **(MARCH 2020)**
2. Mention the differences between interference and diffraction. **(MARCH 2024) (SEP 2021 , 2022)**
3. Give the uses of Polaroids. **(MAY 2022)**
4. State and prove Brewsters law. **(JUNE 2023)**
5. A diffraction grating consist of 4000 slits per centimeter. It is illuminated by a monochromatic light. The second diffraction maximum is produced at an angle of 30°. What is the wavelength of light used. **(JUNE 2024)**
6. Obtain the relation between phase difference and path difference. **(MARCH 2025)**
7. Write a short note on Nicol prism. (**JULY 2025**)

8. DUAL NATURE OF RADIATION & MATTER

1. Derive an expression for de-Broglie wavelength of electrons. **(MARCH 2020)**
2. What are the characteristics of photons? **(SEP 2020 , 2021, JUNE 2024)**
3. List out the laws of photo electric effect. **(MAY 2022, MAR 2023, JULY 2025)**
4. Give the construction and working of a photo emissive cell. **(SEP 2022)**
5. How many photons per second emanate from a 50 mW laser of 640 nm ? **(MARCH 2024)**

ABS PHYSICS

6. A radiation of wavelength 300 nm is incident on calcium surface. Will photoelectrons be observed? Justify your answer. (Work function of calcium 3.20eV) **(MARCH 2025)**

9. ATOMIC & NUCLEAR PHYSICS

1. What are the important inferences from the average binding energy curve? **(MARCH 2020)**

2. Half lives of two radioactive elements A and B are 20 minutes and 40 minutes respectively. Initially the samples have equal number of nuclei. Calculate the ratio of decayed number of A and B nuclei after 80 minutes. **(SEP 2020)**

3. $_{92}U^{235}$ nucleus emits **2α** particles, **3β** particles and **2γ** particles. What is the resulting atomic number and mass number? **(SEP 2021)**

4. Find the (i) Angular momentum
(ii) Velocity of the electron revolving in the 5th orbit of hydrogen atom.
($h = 6.6 \times 10^{-34}$ Js ; $m=9.1\times10^{-31}$ kg) **(MAY 2022, MAR 2025)**

5. What are the properties of Cathode rays? **(SEP 2022)**

6. Explain the process of Alpha decay with example. **(JUNE 2023)**

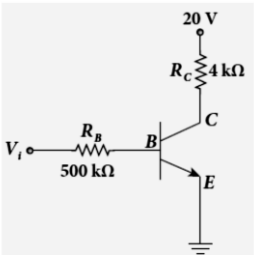
7. Calculate the amount of energy released in joules when 1 kg of $_{92}U^{235}$ undergoes fission reaction..
(MAR 2023) **********

8. Discuss the Beta (B+) decay process with an example. **(MARCH 2024)**

9. The radius of 5th orbit of hydrogen atom is 13.25 A°. Calculate the debroglie wavelength of the electron orbiting in the 5th orbit. **(JUNE 2024)**

10. Explain ß- decay **(JULY 2025)**

10. ELECTRONICS & COMMUNICATION SYSTEMS



1. In the circuit shown in the figure, the input voltage V, is 20 V, V_{BE}=0 V and V_{CE} =0 V, what are the values of IB, IC and β ? **(MARCH 2020)**

2. Modulation helps to reduce the antenna size in wireless communication - Explain. **(MARCH 2020)**

3. Fibre optic communication is gaining popularity among various transmission media. Justify. **(SEP 2020)**

4. Draw the circuit diagram of a full wave rectifier and draw its input and output waveforms. **(SEP 2021)**

5. Draw the circuit diagram of NPN transistor in Common Emitter Configuration. **(MAY 2022)**

6. What is Zener diode ? Mention any two uses of Zener diode. **(SEP 2022)**

ABS PHYSICS

7. The given circuit has two ideal diodes connected as shown in figure below.Calculate the current flowing through the resistance R . **(MAR 2023)**

8. List out the Advantages and limitations of Frequency modulation. **(JUNE 2023)**

9. **Determine the wavelength of the light emitted from LED, which is made up of GaAsP semiconductor, Whose forbidden energy gap is 1.875 eV. Mention the colour of the light emitted (Given $h=6.6\times10^{-34}$ Js).** ********** **(MARCH 2024)**

10. State and prove Demorgans first theorem. **(JUNE 2024)**

11. Distinguish between intrinsic and extrinsic semiconductors. **(MAR 2025)**

12. In a transistor connected in the common base configuration, the current gain $\alpha=0.95$, the emitter current I_E= 1 mA. Calculate the values of collector current Ic and base current I_B .

(JULY 2025)

GOVERNMENT COMMON PUBLIC EXAMINATION – 5 MARKS

1. ELECTROSTATICS

1. State gauss law. Obtain the expression for electric field due to an infinitely long charged wire.

(MARCH 2020, MAR 2025)

2. Explain the effect of dielectric when inserted in parallel plate capacitor when it is disconnected from the battery. **(SEP 2020)**

3. Derive an expression for electric field due to a dipole at a point on its axial line.

(MAR 2024) (SEP 2021)

4. Derive an expression for electrostatic potential due to an electric dipole.

Discuss its special cases**. (MAY 2022 , JUNE 2024 , JULY 2025)**

5. Explain in detail the principle , construction and working of a Van de Graaff generator.

(SEP 2022 , JUNE 2023)

6. (i) State couloumbs law in electrostatics.

(ii) Differentiate couloumbs force and gravitational force**. (MAR 2023)**

ABS PHYSICS

2. ELECTRICITY

1. How the emf of two cells are compared using potentiometer. **(MARCH 2020 , JUNE 2024, MAR 2025)**

2. Obtain the condition for bridge balance in Wheatstone's bridge. **(SEP 2020 , SEP 2022 , JUNE 2023, JULY 2025)**

3. (i) Derive an expression for equivalent resistance in Series network . **(SEP 2021)**

   A copper wire of cross-sectional area 0.5 mm^2 carries a current of 0.2 A. If the free electron density of copper is $8.4 \times 10^{28}\ m^{-3}$ then compute the drift velocity of free electrons.

4. Determine the internal resistance of a cell using voltmeter. **(MAY 2022)**
5. Describe the microscopic model of current and obtain general form of Ohm's law. **(MAR 2023)**
6. Explain the determination of unknown resistance using metre bridge. **(MAR 2024)**

3. MAGNETISM & MAGNETIC EFFECT OF ELECTRIC CURRENT

1. Derive the expression for the force between two parallel, current carrying conductors. Define – Ampere. **(MARCH 2020)**

2. State Amperes circuital law. Derive an expression for magnetic field due to a current carrying long straight conductor using amperes circuital law. **(JUNE 2023) , (SEP 2020)**

3. State Biot Savart law. Deduce the relation for the magnetic field at a point due to an infinitely long straight conductor carrying current. **(SEP 2021, MAY 2022, JUNE 2024,JULY 2025)**

4. Derive the expression for the force on a current carrying conductor placed in a magnetic field. **(SEP 2022)**

5. Explain in detail the principle ,construction, working of cyclotron. **(MAR 2023)**
6. Calculate the magnetic field produced at a point along the axis of the current carrying circular coil. Write down the equation of the magnetic field at the center of the coil using Biot-Savart law. **(MAR 2024)**
7. Derive the expression for the torque on a current carrying coil in a magnetic field. **(MARCH 2025)**

ABS PHYSICS

4. ELECTROMAGNETIC INDUCTION & ALTERNATING CURRENT

1. Show mathematically that the rotation of a coil in a magnetic field over one rotation induces an alternating emf of one cycle. **(MARCH 2020 , JUNE 2024)**
2. Derive an expression for phase relationship between voltage and current in a AC circuit containing only an inductor. **(MARCH 2020 , MAR 2024)**
3. Explain the principle , construction and working of transformer. **(SEP 2020 , MAY 2022)**
4. Derive an expression for impedance , phase angle between the applied voltage and current in a series RLC circuit. **(SEP 2021 , MAR 2023, JULY 2025).**
5. Show that the mutual inductance between a pair of coils is same. **($M_{12} = M_{21}$) (SEP 2022)**
6. Explain the working of single phase A.C generator with neat diagram. **(JUNE 2023)**
7. **(i)** How will you induce an emf by changing the area enclosed by a coil?

   (ii) A circular metallic disc of area 0.03 m² rotates in a uniform magnetic field of 0.4 T. The axis of rotation passes through the centre and perpendicular to its plane and is also parallel to the magnetic field. If the disc completes 20 revolutions in one second and the resistance of the disc is 42Ω, calculate the induced emf between the axis and the rim. **(MARCH 2025)**

6. ELECTROMAGNETIC WAVES

1. Write down Maxwell equations in integral form. **(MARCH 2020 , SEP 2021 , JUNE 2024, JULY 25)**
2. Write down the properties of electromagnetic waves. **(SEP 2020 , SEP 2022 , MAR 2023)**

   (i) The relative magnetic permeability of the medium is 2.5 and the relative electrical permittivity of the medium is 2.25. Compute the refractive index of the medium. **(SEP 2020, MAR 2023)**

   (ii) Compute the speed of the electromagnetic wave in a medium if the amplitude of electric and magnetic fields are 3×10^4 N C^{-1} and 2×10^{-4} T, respectively. **(SEP 2022)**
3. What is spectrum? Explain the types of emission spectrum. **(MAY 2022)**
4. What is spectrum? Explain the types of absorption spectrum. **(JUNE 2023, MAR 2025)**
5. Explain the Maxwell's modification of Ampere's circuital law. **(MAR 2024)**

ABS PHYSICS

6. RAY OPTICS

1. Derive the mirror equation and the equation for lateral magnification.

   (MARCH 2020 , JUNE 2024, JULY 2025)

2. Derive an expression for angle of deviation produced by a prism and thus obtain the expression for refractive index of material of the prism. **(SEP 2020 , MAR 2024)**

3. Obtain lens maker's formula and mention its significance. **(SEP 2020 , SEP 2021 , MAY 2022)**

4. Describe the Fizeau's method to determine the speed light. **(SEP 2022, MAR 2023, 2025)**

5. What is dispersion? Obtain the equation for dispersive power of a medium. **(JUNE 2023)**

7. WAVE OPTICS

1. Discuss about astronomical telescope and magnification. **(SEP 2021)**

2. Obtain the equation for path difference and bandwidth in Young's double slit experiment. **(MAY 2022 , JUNE 2023, JULY 2025).**

3. Explain about compound microscope and obtain the equation for the magnification.

   (SEP 2022, JUNE 2024)

4. Discuss the diffraction at single slit and obtain the condition for n^{th} minimum. **(MAR 2023)**

5. Explain about simple microscope and obtain equation for magnification for near point focusing and normal focusing. **(MAR 2024)**

6. (i) State Brewster's Law.

(ii) What is the angle at which a glass plate of refractive index 1.65 is to be kept with respect to the horizontal surface so that an unpolarised light travelling horizontal after reflection from the glass plate is found to be plane polarised ? **(MAR 2025)**

8. DUAL NATURE OF MATTER AND RADIATION

1. Describe Davisson – Germer experiment which demonstrated the wave nature of electrons.

   (MARCH 2020 , MAR 2023, JULY 2025)

2. Obtain Einstein's photoelectric equation with necessary explanation. **(SEP 2020 , SEP 2022, MARCH 2024)**

ABS PHYSICS

(ii) Explain the Characteristics of Photon. **(SEP 2022) (JUNE 2023)**

3. Calculate Debroglie Wavelength of electron. **(SEP 2021) (JUNE 2023)**

Calculate the de Broglie wavelength of an electron with kinetic energy 2 eV.

4. Explain Characteristic X-Ray Spectra. **(MAY 2022)**

Calculate the cut-off wavelength and cut-off frequency of x-rays from an x –ray tube of accelerating potential 20,000 V.

5. Derive an expression for de Broglie wavelength of matter waves and electron. **(JUNE 2024)**

6. Explain the construction and working of photo emissive cell and give any two applications of photo cell. **(MAR 2025)**

9. ATOMIC & NUCLEAR PHYSICS

1. (i) Derive the energy expression for an eletron in the n-th orbit hydrogen atom using Bohr atom model. **(MARCH 2020)**

(ii) Calculate Angular momentum of electron with energy -3.4 eV **(MARCH 2020)**

2. Derive the expression for radius of n-th orbit and velocity of the electron using Bohr atom model. **(SEP 2020 , MAR 2023)**

3. State Radioactivity law. Derive an expression for the law of radioactivity.

(SEP 2021 , JUNE 2023, MAR 2025)

4. Explain the Spectral series of Hydrogen atom. **(MAY 2022, JULY 2025)**

5. (i) Define Half life and Mean life. **(SEP 2022)**

(ii) Calculate the number of nuclei of carbon-14 undecayed after 22,920 years if the initial number of carbon-14 atoms is 10,000. The half-life of carbon-14 is 5730 years.

6. What is nuclear reactor? Write note on Moderators, Control rods and Cooling system. **(MAR 2024)**

7. Explain J.J Thomson Experiment to determine the specific charge of electron. **(JUNE 2024)**

10 . ELECTRONICS AND COMMUNICATION

ABS PHYSICS

1. Explain the working of a Transistor as an Oscillator. **(MARCH 2020)**
2. Explain the working of a Transistor as an Amplifier with neat circuit diagram. Draw the input and output Waveforms. **(SEP 2020 , MAR 2023)**
3. What is Frequency Modulation.? List out the Advantages and Limitations of Frequency Modulation. **(SEP 2021 , MAR 2024)**
4.. Explain the construction and working of a full wave rectifier. **(MARCH 2022, JULY 2025)**
5.. State and prove De Morgan's first and second theorem. **(SEP 2022, JUNE 2023, MAR 2025)**
6. What is rectification ? Explain the construction and working of a half wave rectifier. **(JUNE 2024)**

ABS PHYSICS